Capstone Weekly Progress Report

ChalkStream (Group #62)

[Week 1, Oct-29, 2025]

Supervisor: Prof. Safaa Bedawi

Group Members: Cameron Bodkin-Dick, Eric Cui, Sandy Alzabadani

Group meeting dates: Friday, Oct 31st, 10-11 am | Monday, 3 Nov, 10-11 am

1. Overview of Current Week

Began development of the Frame Preprocessing and Content Segmentation components. This includes applying grey scaling, noise smoothing, stroke infilling, and dynamic obstruction removal to frames. As well as creating the v1 prototype of the candidate region detection algorithm for picking up grouped regions of chalk strokes, and setting up for scoring these regions as specific content types with text confidence %.

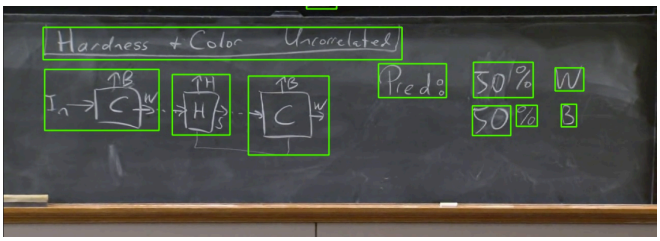
2. Milestone Progress Summary

Component	Description	Current Status % Complete	Remarks / Issues
3.1 Frame Preprocessing	Isolating the board from the rest of the wall, rectifying any angular distortion, and handling dynamic obstructions (instructor walking across the board)	50%	
3.2 Content Segmentation	Determine candidate areas of written content, and assign tex, equation, and diagram affinity	In progress, ~35%	Candidate region detection is in a good spot, fast, accurate, and highly configurable. Want to explore more configuration options with PaddleOCR and performance tests with a CUDA GPU.
3.3. Handwriting OCR	Setting up the testing & training environment for the OCR and prepping datasets.	10%	

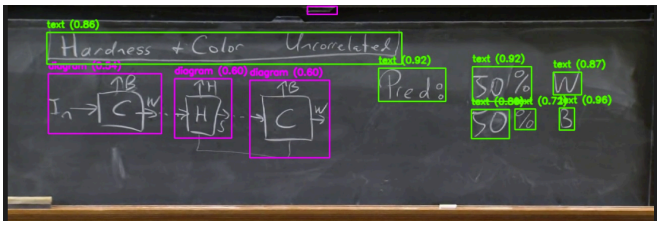
3. Individual Member Contributions

<u>Cameron Bodkin-Dick</u>

This Week: Began developing the basic functionalities of the content segmentation step, which includes creating an algorithm for identifying candidate regions of written content, grouped by proximity. Performed some initial testing with sample images of chalkboard content from LectureVideoDB.



Incorporated the PaddleOCR v5 server and v5 mobile recognition models into the component to enable the scoring of candidate regions. Uses crops from the source image to assign confidence scores and content types for routing into one of the three recognition components.



Next Week: Will continue to explore and finalize the configuration options for the candidate region detection, Paddle recognition settings, and the score/assignment algorithm. Also want to begin performance analysis using a NVIDIA CUDA GPU to get the baselines for processing time across the components with the various mobile and server packages.

Challenges / Dependencies: Progressing fine with cherry-picked data from LectureVideoDB, so there is no rush on image pre-processing (and instructor obstruction removal), but once it is complete, we can work on integrating with Content segmentation and working on live service protocols/logic.

<u>Eric Cui</u>

This Week: Created prototype for the obstructor removal script. The script currently takes in live camera feed or mp4 files and filters out people from the video. The script aim to filter out any human pixels in the video in the form of setting them to the color of perfect black (#000000). The current prototype's goal is to minimize false negative filters, while false positives are only kept in a reasonable range. The obstructor removal has two layers of filters: human filters and movement filters. The two layer filter allows the script to filter out professors much more consistently while keeping false positives reasonable.

Next Week: Improve the obstructor script such that filtered pixels use the previous frame as references, allowing the video to only show the board. Start on the board resize script.

Challenges / Dependencies: There are numerous parameters in the two filters. It is essential to tweak each of the parameters to the right value in order to have the filter applied correctly to match the goal.

<u>Sandy Alzabadani</u>

This week: Worked on creating the environment for handwritten text detection. Ran into some technical issues setting up openCV and python on VS Code as some extensions were not correctly installed. I gathered some more datasets for handwritten datasets to accommodate different writing styles.

Next Week: Have the model working and running for the handwriting OCR. Have a prototype ready to present for the next meeting.

Challenges / Dependencies:

Technical issues while setting up the training environment (OpenCV and python) on VS code for Handwritten OCR.